How Early is Early Enough to Have Native-like Prosodic Phrase Boundaries?

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Prosodic phrase boundaries (PPB) can disambiguate otherwise ambiguous utterances.[1] For example, if there is a PPB (#), *chocolate cookies* is two NPs (*chocolate # cookies*) and if there isn't, it is one NP (*chocolate cookies*). Languages differ in how PPBs are marked, and English PPBs are primarily indicated by pre-boundary lengthening and longer "pausing" at the boundary.[1] People who learn an L2 when they are young often have native-sounding accents, but it's unclear how young is young enough. We investigated how early language background affects PPBs.

Participants. Participants were 24 native English (NE) -speaking adults with two NE-speaking parents and 39 adults with no NE-speaking parents (henceforth Early ESL speakers) who said English was their best language. By report, 23 Early ESL speakers learned English from birth and 16 learned English in early childhood. No participants had a hearing or language impairment.

Procedure. In the comprehension task, participants listened to 102 prerecorded sentences that described 2 items (e.g., *chocolate cookies and milk*) or 3 items (*chocolate # cookies and milk*) and chose which of two pictures matched (Fig 1). A naïve NE speaker said the sentences. In the production task, participants described these 102 pictures using the frame, "I see the _ _ and _ in this picture." Both tasks used 2- and 3-item versions of the same 51 minimal pairs of sentences.

Comprehension results. Comprehension accuracy and RTs were analyzed using 2 (NE/Early ESL) × 2 trial type (2- or 3-item) Bayesian ANOVAs. The best-fitting model for accuracy (BF_{10}= 8.32 x 10^{+14}) included only language background, with NE speakers being 5.3% more accurate than Early ESL speakers (BF_{inc} = 1.20x10^{+15}). The best-fitting model for RTs was the null model.

Production results. We analyzed productions that were 9 or 10 words long (the word count of "I see (the) _ _ and _ in this picture"). Accuracy was determined using Whisper, OpenAI's speech-to-text (STT) model, which was fine-tuned to detect PPBs. If Whisper inserted the correct number of commas (0 for 2-item trials, 2 for 3-item trials), the trial was deemed correct. Two trained RAs who were blind to trial type and participants' language background annotated 200 trials. Human inter-rater and SST-human rater concordances were 100%. For accuracy, a 2x2 ANOVA revealed the best-fitting model (BF_{10} = 1.99x10^{+5}) included language background, trial type, and their interaction. NE speakers were 3.0% more accurate than Early ESL speakers (BF_{inc}= 9.46x10^{+3}), and speakers were 3.5% more accurate on 2- than 3-item trials (BF_{inc} = 2.63x10^{+2}), with NE speakers only outperforming Early ESL speakers on 3-item trials (BF_{inc} = 1.12x10^{+2}, Fig 2). For word duration, a 2 (NE/Early ESL) x 2 (trial type) x 4 critical words (Noun1, Noun2, And, Noun3) ANOVA revealed that the best-fitting model (BF_{10} = ∞) included language background, trial type, word, and the word x trial type and language background x duo interactions. NE speakers' words were 6.8% shorter (BF_{inc} = 1.49×10^{+12}), words in 3-item trials were 22.2% shorter (BF_{inc} 1.49×10^{+12}), and word lengths differed as shown in Fig 3 (BF_{inc} = 1.49×10^{+12}). The word x trial type interaction was due to Noun1 and Noun2 shorter in 2-item than 3-item trials (BF_{inc} = 9.01×10^{+12}, Fig 4), and the language background x trial type interaction was due to the NE-Early ESL difference being greater for 3-item than 2-item trials (BF_{inc} = 1.85x10^{+4}, Fig 5).

Discussion. Neonates recognize the prosody of their mother's language,[2,3] and students, who were exposed to an L1 as children but no longer speak or understand it, sound more native-like after a year of college instruction in the L1 than students without such childhood exposure.[4] Prepubertal L2 exposure is often claimed to be necessary and sufficient for native-like fluency, but some studies suggest native-like fluency is attainable after puberty,[5] and others argue that prepubertal L2 exposure doesn't guarantee it.[6] Our results indicate very early exposure isn't enough: even adults who learned English from birth or early childhood and said English was their dominant language but whose parents weren't NE speakers understood and said PPBs differently than NE speakers with NE-speaking parents. More broadly, the results suggest that, for prosody, it's not just the age of language exposure, but also the nature of language exposure, that matters.

Two-item trial audio: “Choose the side with the chocolate cookies and milk now.” (right side matches)
Three-item trial audio: “Choose the side with the chocolate # cookies and milk now." (left side matches)

Fig. 1. Sample Comprehension Trial: Participants listened to the audio and chose the matching side

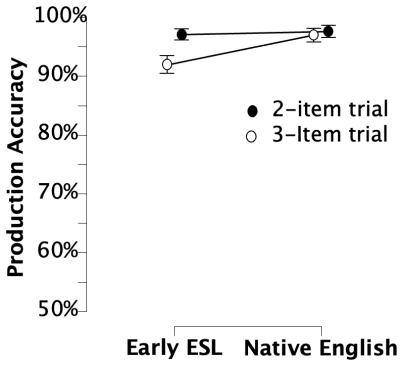

Fig. 2: Production Accuracy*

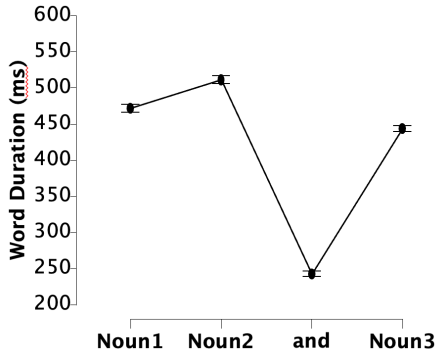

Fig. 3: Critical Words Duration*

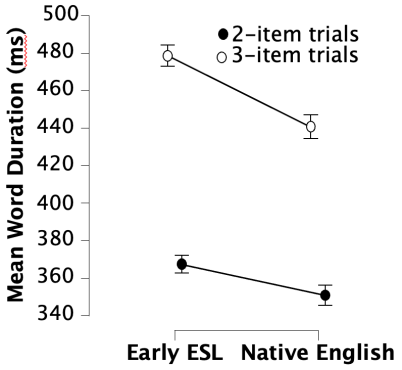

Fig. 4: Duration: Language Background x Trial Type

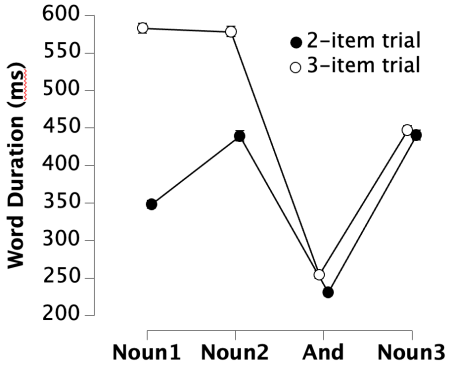

Fig. 5: Duration: Critical Word x Trial Type*

* Error bars are 95% credible intervals

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